

Explanation Of The Script

Before understanding the code, we should first consider a generic approach to the problem. The code deals with an array with a long list of numbers and our aim is to find the maximum and minimum values from that list.

The detailed form of the array is mentioned above. It is to be noted that the first number within the array always starts from the 0th element and the number of elements is always denoted by n-1.

```
maximumValue = function (array)
max = array[0];
```

In this line of code a new function known as the 'maximumValue' is declared that accepts an argument called array. Next a variable called mx is declared with an initial value which is the first value of the array. It is to be assumed that the first value with the array is a temporary maximum value. For instance if the initial value of an array is zero and the largest number within the array is a negative number, then the initial value declared within the array shall be considered as the maximum value.

```
for (i=0; i<array.length; i++)
```

The next part of the code deals with the for loop. Here a simple loop is initiated that counts from the value zero and finally reaches the end of our array. The statement i=0 initializes the variable i. The next part of the loop i<array.length makes the counter progressively pass through each value in the array until the end is reached.

```
if (array[i]>max) {
max = array[i];
}
```

In the above lines of code, the number within the array is checked at the ith position. If the number in this position is greater than the maximum value, the current value is then set as the maximum value. The next statement returns mx as the maximum value of the array which is stored in mx.